

Sovereign Sentinel Architecture V1.0

A Proposed Multi-Tier Control Framework for Frontier AI Safety
Technical Proposal

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Author's Note

This document proposes a comprehensive safety architecture for frontier AI systems. It builds on substantial prior work in mechanistic interpretability, activation monitoring, constrained optimization, and defense-in-depth security principles. While the specific five-axis integration and latency-optimized design represent novel contributions, the individual components draw heavily from existing research.

Critically, this architecture has not been empirically validated at scale. The performance specifications (latency bounds, detection rates, false positive rates) represent engineering targets based on theoretical analysis and small-scale feasibility studies, not production benchmarks. Rigorous empirical validation would require resources and infrastructure available only at Tier 1 research laboratories.

This proposal seeks collaborative partnership with an established AI safety team to conduct proof-of-concept implementation and adversarial evaluation of at least the foundational axes (CLR-CRAE and FSA-HI) before considering production deployment.

Executive Summary

The Sovereign Sentinel Architecture (SSA) is a proposed defense-in-depth framework designed to address two specific failure modes in frontier model deployments: Safety Amnesia (gradual erosion of safety constraints across extended context windows) and Stochastic Sabotage (incremental assembly of prohibited knowledge through individually benign queries).

The architecture proposes replacing stateless linguistic safety prompts with a five-axis hardware-software control stack. These axes operate at fundamentally different abstraction levels—mathematical (training-time weight geometry), computational (inference-time activation trajectories), electronic (hardware interrupt logic), cryptographic (session identity verification), and institutional (human-in-the-loop arbitration).

A core engineering constraint governs the design: no safety mechanism may occupy the synchronous token generation critical path unless physically necessary. This yields a theoretical per-token overhead of 0.85ms (0.8ms for buffered activation probing + 0.05ms for hardware state machine transitions), with heavier statistical computations relegated to asynchronous inter-turn windows.

The central thesis is that no single enforcement mechanism provides adequate safety assurance for frontier models. The SSA is designed so that defeating any axis requires simultaneously bypassing mechanisms operating at multiple abstraction levels—a significant increase in attack complexity compared to single-layer defenses.

Relationship to Prior Work

This architecture builds directly on several lines of recent research:

Mechanistic Interpretability and Activation Monitoring

The CLR-CRAE probe ensemble (Axis 1) extends activation probing techniques developed by multiple research groups. Templeton et al. (2024) demonstrated that sparse autoencoders can learn interpretable features from language model activations, forming a semantic dictionary that captures prohibited reasoning patterns. Our heterogeneous probe architecture combines three probe types—linear boundary classifiers, SAE feature detectors, and Mahalanobis distance anomaly detection—to achieve certified robustness against adversarial activation steering.

The certified robustness requirement addresses the concern raised by Casper et al. (2024) regarding the brittleness of single-probe systems: a sophisticated adversary may encode prohibited reasoning in activation subspaces orthogonal to a linear classifier's decision boundary. By requiring two-of-three probe agreement, we aim to substantially increase the difficulty of evasion.

References: Templeton et al., "Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet" (2024); Marks et al., "Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models" (2024); Casper et al., "Black-Box Access is Insufficient for Rigorous AI Audits" (2024); Li et al., "Inference-Time Intervention: Eliciting Truthful Answers from a Language Model" (2023).

Constrained Training and Lagrangian Methods

The Lagrangian formulation in the CLR-CRAE training phase draws from established constrained optimization theory and recent applications to safe reinforcement learning. Achiam et al. (2017) introduced Constrained Policy Optimization (CPO), which uses dual ascent to enforce safety constraints during policy learning. We adapt this framework to supervised fine-tuning of language models, encoding safety constraints as Lagrangian penalties on prohibited semantic zones in weight space.

The key insight—that training-time weight constraints must be re-instantiated at inference time—addresses a limitation Perez et al. (2022) identified: adversarial prompts can navigate a model to prohibited outputs even when weights remain within the constrained region. Our architecture enforces the same constraint structure on both weights (training) and activations (inference).

References: Achiam et al., "Constrained Policy Optimization" (2017); Ray et al., "A Comprehensive Survey of Constrained Reinforcement Learning" (2019); Perez et al., "Red Teaming Language Models with Language Models" (2022).

Defense-in-Depth and Circuit Breakers

The multi-layer philosophy follows established security engineering principles. Saltzer and Schroeder (1975) articulated defense-in-depth as a core principle in secure system design, arguing that security mechanisms at different abstraction levels provide redundancy against correlated failures.

The hardware interrupt mechanism (Axis 2: FSA-HI) is conceptually related to "circuit breaker" proposals by Anthropic (2024) and the refusal training work by Ganguli et al. (2022). However, our implementation differs in two key aspects: (1) the FSA operates on a dedicated FPGA co-processor, placing it below the software abstraction layer where prompt-based attacks operate, and (2) the Non-Maskable Interrupt architecture ensures the termination decision cannot be overridden by any software component, including the primary model.

References: Saltzer & Schroeder, "The Protection of Information in Computer Systems" (1975); Anthropic, "Core Views on AI Safety" (2023); Ganguli et al., "Red Teaming Language Models to Reduce Harms" (2022); Soares et al., "Supervising strong learners by amplifying weak experts" (2021).

Session-Level Monitoring and Behavioral Drift Detection

The Bayesian Weaver (Axis 4) addresses what Perez et al. (2022) termed "many-shot jailbreaking": attacks that assemble prohibited capabilities across extended interactions through individually benign queries. Our approach uses KL-divergence monitoring of SAE feature distributions to detect both abrupt shifts and slow-drift attacks.

The cumulative drift integral addresses a specific vulnerability: adversaries who understand per-turn thresholds might design attacks that remain just below the threshold on every turn while accumulating substantial total divergence. This technique is inspired by anomaly detection in cybersecurity, particularly the work on slow-rate DDoS attacks by Xiang et al. (2004).

References: Perez et al., "Discovering Language Model Behaviors with Model-Written Evaluations" (2022); Wei et al., "Jailbroken: How Does LLM Safety Training Fail?" (2023); Xiang et al., "Detecting DDoS Attack Traffic at Network Border or ISP" (2004).

Human-in-the-Loop Systems and Expertise Verification

The Einstein Exception mechanism (Axis 5) draws from research on controlled access to dual-use information. Brundage et al. (2018) argued that blanket prohibitions on disseminating potentially dangerous information create capability bottlenecks for legitimate research. The challenge is distinguishing genuine experts from adversaries.

Our ZKP-ETV-HOA protocol (Axis 3) attempts to address this through zero-knowledge proof verification with human-origin attestation. The ZKP component builds on work by Goldreich et al. (1991) on interactive proof systems. The human-origin attestation layer—using keystroke dynamics and hardware-attested input—addresses the emerging threat of LLM-assisted credential gaming, as discussed by Solaiman et al. (2023).

References: Brundage et al., "The Malicious Use of Artificial Intelligence" (2018); Goldreich et al., "Proofs that Yield Nothing but their Validity" (1991); Solaiman et al., "The Gradient of Generative AI Release: Methods and Considerations" (2023); Kang et al., "Continuous Authentication Using Mouse Dynamics and Neural Networks" (2000).

Novel Contributions

While building on the above work, this architecture makes several novel contributions:

1. Integration of five mechanisms at fundamentally different abstraction levels (mathematical, electronic, cryptographic, statistical, institutional) into a unified framework
2. One-token buffered parallelism architecture that enables rigorous activation-level verification with minimal latency overhead
3. Separation of epistemic trust (expertise verification) from intentional trust (behavioral monitoring) with explicit acknowledgment that expertise does not imply benign intent
4. Cumulative drift integral for detecting slow-rate session-level attacks that defeat per-turn thresholds
5. Formally verified rule-based arbitration engine to avoid using an LLM to evaluate LLM outputs (preventing correlated failures)

Threat Model

The SSA is designed against three adversary classes, listed in increasing order of sophistication:

1. Prompt-Level Adversary

Manipulates the model's input context to elicit prohibited outputs without modifying model weights. Current jailbreak techniques (Wei et al., 2023), persona injection, many-shot steering attacks (Perez et al., 2022), and direct activation steering via soft prompts fall into this category. This adversary is assumed to have black-box query access to the deployed model.

2. Session-Level Adversary

Operates across extended interactions, assembling dangerous capabilities through sequences of queries that are individually benign but collectively constitute prohibited knowledge transfer. This adversary is specifically designed to defeat single-turn safety filters and is assumed to have read this proposal in full, including all published threshold values. The attack strategy involves either abrupt shifts (large per-turn divergence) or slow-drift attacks (small per-turn changes accumulating to substantial total divergence).

3. Deceptive Model Adversary

The most sophisticated threat: a model that has learned to produce compliant outputs under evaluation conditions while routing toward misaligned objectives during deployment (Hubinger et al., 2021). This

adversary may have internalized knowledge of the Lagrangian constraint structure during training and may attempt to encode prohibited reasoning in activation subspaces orthogonal to the probe classifiers' trained decision boundaries.

This threat motivates the certified-robust heterogeneous probe ensemble in Axis 1 and the one-token buffering architecture that ensures probe evaluation cannot be bypassed by exploiting timing relationships between generation and verification.

All three adversary classes are assumed to have read this proposal. The security argument does not depend on obscurity of design. It depends on the computational and institutional cost of simultaneously defeating mechanisms operating at four distinct abstraction levels.

Reference: Hubinger et al., "Risks from Learned Optimization in Advanced Machine Learning Systems" (2021).

Latency Architecture: Core Principles

Before specifying individual axes, we state the production latency strategy explicitly, as it governs the implementation of every component. These principles aim to make safety monitoring compatible with production service level agreements (SLAs) while maintaining rigorous enforcement.

Principle 1: Critical Path Minimization

Only the FSA hardware interrupt and the one-token emission buffer are permitted to occupy the synchronous token generation critical path. All other safety mechanisms are assigned to parallel compute lanes or asynchronous inter-turn windows. This is not a performance optimization; it is an architectural constraint enforced at the hardware level.

Principle 2: One-Token Buffered Parallelism

The CLR-CRAE probe ensemble evaluates activations from token N while token N+1 is being generated speculatively on the primary inference engine. Token N+1 is held in an emission buffer and not released to the user until probe evaluation of token N completes (targeted at 0.8ms maximum). The probe co-processor and primary inference engine operate on separate compute lanes with dedicated memory bandwidth allocation.

If a probe violation is detected on token N, the halt signal prevents token N+1 from being emitted. The effective safety window is exactly one token—formally sufficient for any realistic prohibited output sequence, which requires multiple tokens to become actionable.

This architecture sacrifices the theoretical zero-latency parallelism of unbuffered speculative execution in favor of a physically achievable one-token safety guarantee. The 0.8ms buffering cost per token represents the honest engineering price of rigorous activation-level verification.

Principle 3: Inter-Turn Asynchrony for Session Monitoring

The Bayesian Weaver operates at turn granularity. A turn completes when the model finishes generating its response. Weaver evaluation runs asynchronously during the interval between response delivery and the user's next input—a window that in human interaction is measured in seconds and in automated pipelines would be enforced as a mandatory 10ms inter-turn buffer. Weaver computation never touches the token generation critical path.

Principle 4: Event-Driven Escalation

The CCE arbitration engine activates only on escalation events, which should be rare by design. It is never in the normal inference critical path. Its computational cost is irrelevant to steady-state latency.

Projected Production Latency Overhead

Under this architecture, user-perceived latency overhead would theoretically be 0.85ms per token in steady state—0.8ms for buffered probe evaluation and 0.05ms for the FSA transition cost. For a 500-token response, total overhead would be approximately 425ms, representing less than 5% of total inference time for a 70B model on H100 SXM5 hardware at typical serving throughput (40-60 tokens/second).

Important caveat: These are engineering targets based on theoretical analysis, not empirically validated benchmarks. Actual performance would need to be measured under realistic serving conditions with proper hardware configuration.

Phase 1: The Trident Protocol (Years 1-2)

Thesis: Establishing Deterministic Boundaries via Hardware-Software Co-Design

The Trident Protocol constitutes the SSA's foundational layer: safety invariants enforced mathematically during training via Lagrangian optimization, extended into a certified-robust runtime enforcement regime running on a buffered parallel co-processor, and interrupted electronically by dedicated FPGA hardware when violated.

Axis 1: Constitutive Lagrangian Regularization with Certified-Robust Activation Enforcement (CLR-CRAE)

The Lagrangian Foundation—Training Phase

Safety invariants would be injected into the model's weight space during training and fine-tuning as Lagrangian constraints on the optimization objective. The constrained loss is formalized as:

$$L(w, \lambda) = f(w) + \sum_i \lambda_i g_i(w)$$

where:

- $f(w)$ is the standard task loss
- $g_i(w)$ are constraint functions encoding prohibited semantic zones in weight space
- λ_i are Lagrange multipliers enforced via dual ascent

The dual ascent update at each training step would be:

$$\lambda_i \leftarrow \max(0, \lambda_i + \alpha_i g_i(w))$$

where α_i is the per-constraint step size, tuned to ensure convergence to a stable saddle point. At convergence, the Karush-Kuhn-Tucker (KKT) conditions would be satisfied: the model's weight distribution would be mathematically biased against prohibited semantic zones, with safety constraints satisfied as a property of the weight geometry itself rather than as a post-hoc filter.

The constraint functions g_i would be defined over a structured taxonomy of prohibited action classes—proposed as three top-level domains (CBRN synthesis, critical infrastructure exploitation, and mass-targeting sequences) with a minimum of 40 sub-categories per domain. Each g_i would be constructed from probe classifiers trained on activation data from known prohibited outputs.

Why the Lagrangian Framework Must Extend Into Runtime

The Lagrangian formulation provides a rigorous guarantee about weight geometry at the conclusion of training. When the KKT conditions are satisfied, no unconstrained descent direction exists in weight space that simultaneously reduces task loss and violates a safety constraint. This is formally verifiable and would constitute the architecture's strongest mathematical guarantee.

However, training-time constraints govern the weight distribution, not the inference-time activation trajectory. A capable adversary could potentially navigate through a compliant weight space to prohibited output regions via soft prompting, chain-of-thought scaffolding, or activation steering—

reaching semantically prohibited destinations without the weights leaving the Lagrangian feasible region.

Therefore, the same constraint structure enforced on weights during training must be re-instantiated as a live verification predicate on activations during inference. This is not a different safety mechanism; it is the same Lagrangian framework operating at a different point in the forward pass.

Runtime Enforcement—The One-Token Buffered Parallel Co-Processor

The CLR-CRAE extension would re-evaluate the Lagrangian safety invariant against residual stream activations at every forward pass. This evaluation would run on a dedicated co-processor lane operating in buffered parallel with the primary inference engine.

One-Token Buffered Parallelism Architecture:

The probe co-processor would receive a copy of token N's residual stream activations via a dedicated memory bus while token N+1 is generated speculatively on the primary inference engine. Token N+1 would be held in an emission buffer and not released until probe evaluation of token N completes (targeted 0.8ms maximum). The probe co-processor evaluates the probe ensemble against token N's activations and issues a pass or halt signal.

- Normal case: The signal is pass, token N+1 is released from the emission buffer after 0.8ms, and generation continues.
- Violation case: The halt signal is issued, token N+1 is discarded from the buffer before emission, and the session is terminated or escalated. The effective safety window is exactly one token—token N+1 never reaches the user.

This architecture aims to ensure the one-token safety guarantee is physically achievable. The 0.8ms buffering cost per token represents the honest engineering price of rigorous activation-level verification.

Buffer Overflow Handling

If probe evaluation queue depth exceeds 2 tokens (indicating abnormal hardware contention), the system would enter fail-safe mode: generation pauses until the queue clears. This would be a rare degraded-performance state, not a safety violation. Target median recovery time: <50ms. Proposed deployment alert threshold: >0.01% of inference sessions.

Persistent buffer overflow (>5 consecutive occurrences) would trigger a mandatory hardware diagnostic review before inference resumes.

The Heterogeneous Probe Ensemble

A heterogeneous probe ensemble would be evaluated at monitored layers $\{L/4, L/2, 3L/4\}$, where L is the total layer count. Three independently trained probe types would operate on residual stream activation at each monitored layer:

Probe Type A—Linear Boundary Probe:

Directly extends the training-time Lagrangian constraint into activation space. For activation a_l at layer l :

$$\hat{g}_l(a_l) = w_l^T a_l + b_l \leq \epsilon_l$$

Probe Type B—SAE Feature Dictionary Probe:

Operates in the sparse autoencoder feature basis (building on Templeton et al., 2024), detecting prohibited-zone proximity in the learned semantic feature space:

$$\hat{g}_l^{SAE}(a_l) = \sum_k \alpha_k f_k(a_l) \leq \epsilon_l^{SAE}$$

where $f_k(a_l)$ are the top-k activated SAE dictionary features at layer l .

Probe Type C—Mahalanobis Distance Anomaly Detector:

Flags activations falling outside the empirical covariance structure of the benign training distribution, independent of any known prohibited cluster:

$$M(a_l) = (a_l - \mu)^T \Sigma^{-1} (a_l - \mu) \geq \tau_m$$

where μ and Σ are the mean and covariance of residual stream activations on the benign training corpus at layer l . This probe aims to capture novel attack strategies routing prohibited reasoning through feature combinations absent from training data.

The ensemble would report a violation if any two of the three probe types exceed their respective thresholds at any monitored layer. A single-probe violation would produce an elevated-risk flag routed to the session monitor. A two-probe violation would issue a halt signal to the primary inference engine.

Certified Robustness Requirement

Because the Deceptive Model Adversary may encode prohibited reasoning in an activation subspace orthogonal to the Type A probe's linear decision boundary, the ensemble must satisfy a formal certified robustness bound:

For all perturbations δ with $\|\delta\|_2 \leq \rho$ applied to the residual stream activation, the ensemble should produce a violation flag with probability $\geq 1 - \alpha$.

Proposed deployment prerequisites: $\rho = 0.05$ (ℓ_2 -norm, BF16 precision), $\alpha \leq 0.05$, validated on a held-out adversarial corpus of 50,000 steering-perturbed forward passes per prohibited domain. Any deployment using FP8 quantized inference would require separate robustness evaluation under FP8 activation statistics, as the certified bound is precision-specific.

Implementation Specifications

- Probe ensemble would run at BF16 precision on dedicated co-processor lane with isolated memory bandwidth allocation
- Probe ensemble would be retrained quarterly using the same dual ascent procedure as the training-time Lagrange multiplier update, maintaining consistency between weight-space constraint geometry and activation-space enforcement boundary
- Probe staleness beyond 90 days would constitute a compliance failure requiring re-certification
- Target false-positive rate: $\leq 0.1\%$ on a held-out benign corpus of 10 million tokens, validated at each quarterly retraining cycle

Latency Profile

- Per-token buffering cost: 0.8ms (target)
- Buffer overflow recovery: <50ms median (target for rare degraded-performance state)

Axis 2: Finite Safety Automaton with Hardware Interrupt (FSA-HI)

Mechanism

A dedicated FPGA co-processor would run a Finite Safety Automaton—a formally specified directed graph $G = (V, E)$ with $|V| \leq 2^{16}$ states—in true parallel with the primary inference pipeline. Each state would represent a semantically meaningful action category derived from the prohibited action taxonomy. The FSA would evaluate the output token stream in real time, updating its state on each emitted token with $O(1)$ per-token computation: a single lookup table transition on the current state and the incoming token's semantic category.

The FSA is the one axis that would be intentionally synchronous with token emission. This is a deliberate design choice: the hardware interrupt must be able to sever the inference pipeline on the current token, not the next one. The target latency cost of $\leq 0.05\text{ms}$ per token at 500MHz FPGA clock would represent the second unavoidable contribution to user-perceived overhead, totaling approximately 25ms over a 500-token response.

Hardware Interrupt Logic

When the FSA enters any state $s \in \text{Sprohibited}$, the FPGA would issue a Non-Maskable Interrupt (NMI) that severs the inference pipeline independently of any software-layer decision. This interrupt could not be masked, overridden, or delayed by the primary model or any software component. Target NMI issuance latency from state-entry to interrupt assertion: $\leq 10\mu\text{s}$, which would be hardware-verified via TPM timestamp delta.

The fail-closed condition would trigger when the FSA enters any state with out-degree zero in the permitted subgraph, ensuring the termination decision is formally decidable.

Every FSA state transition, NMI event, and session termination would be recorded as a SHA-3-512 hash chain entry anchored to the hardware TPM, with microsecond-resolution timestamps and a cryptographic commitment to the session's activation state at the time of the event.

Coverage Requirement

The FSA's state graph would target $\geq 94\%$ recall on a red-team evaluation corpus before deployment, with per-domain recall reported separately. The coverage gap would be formally characterized in the deployment safety case with residual risk quantification for each uncovered prohibited behavior category.

Important note: The 94% target explicitly acknowledges a 6% coverage gap. Any deployment would require formal characterization of which prohibited behaviors fall outside the FSA's state graph.

Latency Profile

- Per-token FSA transition: 0.05ms (target, synchronous with emission)
- NMI issuance latency: $<10\mu\text{s}$ (target, upon state violation)

Axis 3: Cryptographic Identity and Epistemic Trust Verification with Human-Origin Attestation (ZKP-ETV-HOA)

Mechanism

Users requesting access to protected domains would pass a Zero-Knowledge Proof verification protocol confirming domain expertise without disclosing credentials, institutional affiliations, or personal identity. The ZKP challenge library would contain a minimum of 10,000 domain-stratified challenges per protected domain, generated and evaluated by a formally independent co-processor sharing neither weights nor training data with the primary model.

This architectural separation is non-negotiable. A primary model that generated or evaluated its own ZKP challenges would create a pathway for an agentic adversary to submit LLM-generated challenge responses and game the epistemic trust score without possessing genuine domain knowledge.

Human-Origin Attestation Layer

The ZKP co-processor would incorporate a Human-Origin Attestation (HOA) mechanism verifying that challenge responses were generated by an unaided human. At least one sub-challenge per ZKP session would be time-pressured with a response window calibrated to human typing latency (750-2000ms for a 50-token response), with keystroke dynamics recorded against a registered baseline.

ZKP sessions would be submitted via a hardware-attested interface session with TPM-verified human-interface device events, creating a hardware-level provenance record distinguishing physical keyboard input from API calls.

Target HOA deployment prerequisite: $\geq 99\%$ human vs. LLM discrimination on a held-out challenge corpus using keystroke-dynamic timing alone.

The Epistemic Trust Coefficient (τ_e)

Successful ZKP verification with HOA confirmation would produce the Epistemic Trust Coefficient $\tau_e \in [0, 1]$.

Proposed calibration targets:

- Non-expert achieves $\tau_e < 0.3$ with probability ≥ 0.99
- Genuine domain expert achieves $\tau_e > 0.85$ with probability ≥ 0.95
- LLM-assisted submission is rejected by HOA before τ_e is computed, with the rejection event logged to the TPM hash chain

τ_e would be domain-specific and non-transferable. Certificates would be valid for 30 days; re-verification would be required thereafter.

Critical Design Note

The ZKP protocol verifies expertise and human origin. It makes no inference about intent. A high τ_e score indicates that a human possessing domain knowledge submitted the verification. It is not a statement about how that knowledge will be used.

The separation between epistemic trust (τ_e) and intentional trust (τ_i , computed by Axis 4) is a deliberate architectural decision. Conflating the two would create a privilege escalation path: demonstrating expertise would implicitly grant assumed benign intent, which the ZKP protocol has no mechanism to verify.

Latency Profile

- ZKP verification: Asynchronous (pre-session credential establishment)
- Per-token impact: 0ms (credentials verified before inference begins)

Phase 2: The Quad-Axis Framework (Years 3-4)

Thesis: Mitigation of Stochastic Sabotage via Feature-Level Session Monitoring

Phase 2 would add the behavioral monitoring layer addressing session-level adversaries. This layer would operate on the model's internal activations rather than its outputs and is designed around the inter-turn asynchrony principle: all Weaver computation would run in the interval between response delivery and the user's next input, adding zero latency to the token generation critical path.

Axis 4: The Bayesian Weaver—Sparse Autoencoder Session Monitor with Cumulative Drift Enforcement

Mechanism

The Bayesian Weaver would use Sparse Autoencoders (SAEs) to monitor the model's internal activations across the full session trajectory. An SAE of width 32,768 features would be trained on residual stream activations, learning a sparse interpretable dictionary of semantic features underlying the model's representations. The Weaver would evaluate this feature dictionary at each turn, tracking how the session's activation profile evolves.

The Asynchronous Inter-Turn Execution Model

The Weaver would be the computationally heaviest component in the SSA at approximately 4.5ms per turn (target). In a naive synchronous implementation this would add a perceptible delay between every model response and the user's ability to continue. The inter-turn asynchrony principle eliminates this entirely.

The Weaver would receive a copy of the completed turn's activation trajectory at the moment the model finishes generating its response. It would begin evaluation immediately, running on a dedicated compute allocation with no shared memory bandwidth with the primary inference engine. In a human

interaction context, the user is reading the model's response during this window—typically several seconds. The Weaver's 4.5ms computation would complete well within this reading window.

For automated pipeline deployments with no human reading time, a mandatory inter-turn buffer of 10ms would be enforced between response delivery and the next permitted input. This buffer should be imperceptible in any realistic agentic workflow.

If Weaver evaluation is not complete when the next input arrives—which at 4.5ms against a 10ms buffer should not occur in normal operation—the system would apply conservative default trust values, treating τ_i as if it had degraded by one decay step. The system would never default to permissiveness.

KL-Divergence Monitoring

Salami-Slicing attacks would be detected through per-turn KL-divergence monitoring:

$$DKL(P_t || P_0) = \sum_k P_t(k) \log(P_t(k) / P_0(k))$$

where P_0 is the session's baseline feature distribution established in the first three turns and P_t is the feature distribution at turn t . The baseline P_0 would be frozen after turn 3 and not updated, preventing adversarial baseline walking.

Cumulative Drift Integral

Per-turn KL monitoring detects abrupt transitions but is vulnerable to slow-drift attacks that remain below per-turn thresholds while accumulating meaningful total divergence across extended sessions. The Weaver would therefore maintain a session-level cumulative drift integral:

$$\Phi(T) = \sum_{t=1 \dots T} DKL(P_t || P_0)$$

with a session termination threshold $\Phi_{\max}$ calibrated on the red-team corpus. A session whose cumulative integral exceeds $\Phi_{\max}$ would trigger trust degradation regardless of whether any individual turn was anomalous.

Target validation: $\Phi_{\max} \leq 0.5\%$ false-negative rate against slow-drift sessions of 50, 100, and 200 turns.

Novelty Detection

A novelty detection signal would monitor for activation trajectories entering low-density regions of the SAE feature space. Any token whose activation profile falls in the bottom 2nd percentile of the training corpus density estimate would trigger a novelty flag—a trust-degradation signal capturing adversarial strategies routing prohibited reasoning through feature combinations absent from training data.

The Intentional Trust Coefficient (τ_i)

The Weaver would output the Intentional Trust Coefficient $\tau_i \in [0, 1]$, degrading in response to per-turn KL spikes, $\Phi_{\max}$ breach, novelty flags, and prohibited feature cluster proximity:

$$\tau_i(t) = \tau_i(t-1) \cdot \lambda^{\delta(t)}$$

where $\lambda = 0.95$ would be the per-event decay base and $\delta(t)$ would be a severity-weighted event count at turn t . Target calibration: λ on a red-team session corpus of $\geq 5,000$ sessions, achieving a false-negative rate (malicious session reaching $\tau_i > 0.7$) of $\leq 0.5\%$.

Calibration and Distribution Shift Certification

The conformal calibration certificate would require that the 95% prediction interval empirically contain the true intent label with frequency $\geq 94\%$ on the calibration corpus, with a Maximum Mean Discrepancy (MMD) bound:

$$MMD(\text{calibration}, \text{deployment}) \leq \epsilon_{mmd} = 0.08$$

If this bound is exceeded the certificate would be suspended and τ_i -based decisions would revert to conservative defaults. The certificate would be renewed with each quarterly SAE retraining cycle.

Latency Profile

- Per-turn Weaver evaluation: 4.5ms (target, asynchronous during inter-turn window)
- Per-token impact: 0ms (never in critical path)

Phase 3: The Five-Axis Sovereign (Year 5+)

Thesis: Causal Arbitration, Controlled Exception Handling, and Institutional Accountability

Phase 3 would introduce the conflict resolution and exception-handling layer. All Phase 3 components would be event-driven and sit entirely outside the normal inference critical path. Their computational cost would be irrelevant to steady-state latency; what matters is their response latency in escalation scenarios.

Axis 5: The Causal Cross-Examiner—Rule-Based Arbitration Engine with Independent Audit Summarization (CCE-RAE)

Mechanism

The Causal Cross-Examiner would resolve conflicts between the FSA-HI (hardware, deterministic) and the Bayesian Weaver (statistical, probabilistic). It would be implemented as a formally verified rule-based arbitration engine, not an LLM. Using a language model to evaluate whether language model outputs are safe creates a correlated failure distribution that the architecture is explicitly designed to avoid.

The CCE's decision logic would follow a formally specified priority ordering, mechanically verified using an SMT solver (Z3) with a published verification artifact. The ordering would be proven total and acyclic under all reachable input combinations:

6. Hardware FSA interrupt is unconditional and cannot be overridden by any software component, trust coefficient, or exception pathway
7. Two-probe CLR-CRAE ensemble violation requires human review within 90 seconds before inference resumes
8. SAE probe violation above threshold escalates to the session monitor with a 60-second human review window
9. τ_i degradation below 0.4 triggers session termination with a logged justification written to the TPM audit chain

When multiple axes produce simultaneous signals, the CCE would assemble a structured evidence package—containing triggering events, active trust coefficients, activation trajectory summary, FSA state history, $\Phi(T)$ value, and all probe ensemble outputs—and route it to the appropriate response tier. The CCE would classify, package, and escalate. It would not make approval decisions.

Independent Audit Summarization

Evidence packages routed to human reviewers would be accompanied by a natural-language summary generated by a dedicated frozen audit model: a separate system of $\leq 7B$ parameters, trained exclusively on safety audit summarization, with formally distinct weight provenance and no architectural overlap with the primary model.

All audit model summaries would be post-hoc verified by a rule-based consistency checker confirming the summary does not contradict any structured field in the evidence package. Summaries failing consistency verification would be flagged before presentation to reviewers.

The Einstein Exception and the Irreversibility Floor

The SSA recognizes that a system treating all high-risk queries as equivalent regardless of context is a capability limiter rather than a safety system. The Einstein Exception would provide a formal pathway for verified expert users to engage with high-risk material under controlled conditions.

A session achieving $\tau_e > 0.95$ (ZKP co-processor with HOA confirmation) and $\tau_i > 0.88$ (Weaver with valid calibration certificate) could submit a formal exception request. These thresholds would be published; security would derive not from their secrecy but from the difficulty of satisfying both simultaneously through legitimate means.

An approved exception request would initiate a 72-hour human quorum of a minimum of three reviewers from a standing panel with externally verified domain expertise. The quorum decision would be binding, recorded to the TPM hash chain, and subject to retrospective external audit on a 30-day rolling basis.

The Irreversibility Floor:

The Irreversibility Floor would be an absolute limit the Einstein Exception cannot override. Any action classified as irreversible by the FSA's irreversibility sub-graph—currently proposed to include CBRN synthesis routes, critical infrastructure modification sequences, and mass-scale data exfiltration operations—would bypass the exception pathway entirely. No trust coefficient combination would unlock these actions without human quorum. The Irreversibility Floor would be a formally specified property of the FSA state graph, not a policy setting, and would not be modifiable through configuration or runtime instruction.

Latency Profile

- CCE arbitration: Event-driven (activates only on escalation, not in normal critical path)
- Human review window: 60-90 seconds (depending on violation severity)
- Einstein Exception quorum: Target median 54 hours, 95th percentile 72 hours

Axis Interaction Summary

Defeating the SSA would require simultaneously bypassing mechanisms at four distinct abstraction levels:

Mathematical / Activation-Space:

CLR-CRAE would enforce weight geometry via Lagrangian dual ascent during training and re-evaluate the same constraint structure against live residual stream activations on every forward pass using a certified-robust heterogeneous probe ensemble. A deceptive model would need to produce activations satisfying all three probe types across all monitored layers while navigating toward a prohibited output—formally bounded in difficulty by the certified robustness specification at $p = 0.05$.

Electronic:

The FSA-HI would issue a Non-Maskable Interrupt no software component can mask, operating below the reach of prompt-level or session-level adversaries.

Cryptographic / Identity:

The ZKP-ETV-HOA would perform trust verification through a component with no shared representations with the primary model, augmented by hardware-attested human-origin verification that closes the LLM-assisted gaming pathway.

Statistical-Behavioral:

The Bayesian Weaver would monitor trajectory drift using both per-turn KL-divergence and the cumulative drift integral $\Phi(T)$. Slow-drift attacks defeating per-turn monitoring would be captured by $\Phi(T)$; abrupt attacks defeating $\Phi(T)$ would be captured by per-turn monitoring. Novelty detection would capture both.

Institutional:

The CCE-RAE would route conflicts to human reviewers via a formally verified arbitration engine, with independent audit summaries verified by a rule-based consistency checker. Quorum decisions would be externally audited on a 30-day rolling basis.

Proposed Implementation Roadmap and Collaboration Framework

This section outlines a phased approach to validating the SSA's core mechanisms through collaborative research with a Tier 1 AI safety team. The roadmap prioritizes empirical validation of the most critical axes before considering production deployment.

Phase 0: Proof-of-Concept Validation (Months 1-6)

Objective

Demonstrate technical feasibility of Axis 1 (CLR-CRAE) and Axis 2 (FSA-HI) at small scale. Establish baseline performance metrics and identify implementation challenges.

Deliverables

10. CLR-CRAE Prototype:
 - Implement Lagrangian-constrained fine-tuning on a 7B parameter model using 3-5 constraint functions for a single prohibited domain (e.g., CBRN synthesis precursors)
 - Train heterogeneous probe ensemble (linear, SAE-based, Mahalanobis) on model's residual stream activations at three monitoring layers
 - Benchmark probe evaluation latency on single GPU with simulated one-token buffering
 - Measure false positive rate on 100k-token benign corpus and detection rate on red-team corpus
11. FSA Prototype:
 - Design state graph for single prohibited domain with 50-100 states
 - Implement software FSA simulator with $O(1)$ state transitions
 - Benchmark transition latency and recall on red-team corpus
12. Technical Report: Document implementation details, performance measurements, and lessons learned. Publish as arxiv preprint to solicit community feedback.

Success Criteria

- CLR-CRAE achieves $\geq 85\%$ detection rate with $\leq 1\%$ false positive rate on held-out corpus
- Per-token probe evaluation completes within 2ms (relaxed target for software prototype)
- FSA achieves $\geq 80\%$ recall on red-team corpus with $O(1)$ verified transition complexity
- Identified technical blockers have clear engineering solutions

Phase 1: Hardware Co-Processor Integration (Months 7-12)

Objective

Demonstrate that dedicated co-processor hardware can achieve the proposed latency targets (0.8ms for buffered probing, 0.05ms for FSA transitions) while maintaining detection quality.

Deliverables

13. Probe Co-Processor: Implement CLR-CRAE on dedicated GPU with isolated memory bandwidth. Configure one-token emission buffer and measure actual end-to-end latency under realistic serving load.

14. FPGA FSA: Port software FSA to FPGA hardware (Xilinx or Intel). Implement NMI logic and measure interrupt latency with hardware timestamps.
15. Integration Testing: Deploy both co-processors alongside 7B primary model. Measure combined system latency, throughput degradation, and failure modes under contention.
16. Adversarial Evaluation: Work with internal red team to attempt bypass of both axes simultaneously. Document successful attacks and refine probe training.

Success Criteria

- Probe co-processor achieves $\leq 1.2\text{ms}$ per-token overhead (allowing 50% margin on 0.8ms target)
- FPGA FSA achieves $\leq 0.1\text{ms}$ per-token overhead (2x margin on 0.05ms target)
- Buffer overflow frequency $< 0.1\%$ under 95th percentile serving load
- Red team identifies specific attack vectors but not systematic bypass of both axes

Phase 2: Certified Robustness Validation (Months 13-18)

Objective

Validate the certified robustness bound ($p = 0.05$, $\alpha \leq 0.05$) through extensive adversarial perturbation testing. This is critical for defending against the Deceptive Model Adversary.

Deliverables

17. Perturbation Corpus: Generate 50k adversarial activation trajectories per prohibited domain using gradient-based steering, soft prompts, and automated jailbreak techniques.
18. Robustness Certification: For each prohibited output in corpus, compute minimum ℓ_2 perturbation required to evade two-of-three probe detection. Empirically validate that $\geq 95\%$ of successful evasions require $\|\delta\|_2 > 0.05$.
19. Probe Ensemble Refinement: If robustness bound is not achieved, iterate on probe training (constraint function design, SAE architecture, Mahalanobis covariance estimation) until bound is satisfied.
20. Precision Study: Repeat robustness validation under FP8 quantization to assess whether certified bound holds at reduced precision.

Success Criteria

- Certified robustness bound validated at $p = 0.05$, $\alpha \leq 0.05$ for BF16 activations
- False positive rate remains $\leq 0.15\%$ on benign corpus (allowing some degradation from probe refinement)
- FP8 robustness study identifies whether reduced precision deployment requires bound adjustment

Phase 3: Session Monitoring Prototype (Months 19-24)

Objective

Validate Axis 4 (Bayesian Weaver) against session-level adversaries attempting many-shot jailbreaking and slow-drift attacks.

Deliverables

21. SAE Training: Train 32k-feature sparse autoencoder on residual stream activations. Validate reconstruction quality ($\text{MSE} \leq 0.03$) and feature interpretability.
22. Weaver Implementation: Implement per-turn KL-divergence monitoring, cumulative drift integral, and novelty detection. Configure to run asynchronously during 10ms inter-turn buffer.

23. Session-Level Red Teaming: Generate corpus of malicious multi-turn sessions using automated many-shot attacks and human adversarial role-play. Include both abrupt and slow-drift strategies.
24. Calibration: Tune λ decay parameter and $\Phi_{\max}$ threshold to achieve $\leq 1\%$ false-negative rate on red-team session corpus with $\leq 0.5\%$ false-positive rate on benign session corpus.

Success Criteria

- Weaver computation completes within 6ms per turn (allowing margin on 4.5ms target)
- False-negative rate $\leq 1\%$ on session-level red-team corpus
- False-positive rate $\leq 0.5\%$ on 10k benign multi-turn sessions
- Cumulative drift integral successfully detects slow-drift attacks that evade per-turn monitoring

Beyond Phase 3: Full System Integration

Phases 0-3 represent approximately 24 months of focused research and engineering. If these phases successfully demonstrate technical feasibility, subsequent work would address:

- Axis 3 (ZKP-ETV-HOA): Human-origin attestation validation, keystroke dynamics discrimination, challenge library development
- Axis 5 (CCE-RAE): Formal verification of arbitration logic, independent audit model training, human quorum workflow design
- Scale Testing: Deployment on frontier-scale models (70B+) under production serving conditions
- Operational Monitoring: Design and implementation of weekly safety metrics dashboard, alert thresholds, and incident response protocols
- External Audit: Independent security review by third-party AI safety researchers

Required Resources

Successful implementation requires access to infrastructure and expertise beyond individual researcher capacity:

- Compute: Dedicated GPU cluster for probe training and SAE training; FPGA development boards for FSA prototyping
- Models: Access to frontier-scale model weights and training pipelines for Lagrangian-constrained fine-tuning
- Red Team: Experienced adversarial researchers for jailbreak development and session-level attack design
- Expertise: Collaboration with interpretability researchers (SAE training), hardware engineers (FPGA implementation), security researchers (formal verification)
- Evaluation Infrastructure: Large-scale benign corpus for false-positive testing; comprehensive red-team corpus covering multiple attack vectors

Open Questions for Collaborative Research

Several critical questions require empirical investigation:

25. Is one-token buffering sufficient? Could adversarial outputs encode meaningful dangerous information in a single token, or do all realistic threats require multi-token sequences?
26. How does probe ensemble performance scale with model size? Do 70B+ parameter models exhibit more complex activation geometries that reduce certified robustness?
27. Can the cumulative drift integral be gamed by adversaries who understand the $\Phi_{\max}$ threshold? What is the optimal balance between false-positive and false-negative rates for session-level monitoring?
28. How brittle is the full system? If one axis is compromised, does the architecture degrade gracefully or fail catastrophically?

29. What is the operational cost-benefit tradeoff? Does the increased safety assurance justify the implementation complexity and maintenance burden?

Known Limitations and Open Problems

The SSA does not claim to solve AI alignment. It aims to make adversarial circumvention of frontier model safety constraints significantly harder, measurably detectable, and institutionally accountable. Four limitations are explicitly acknowledged:

1. SAE Feature Coverage

The feature dictionary is trained on known prohibited-zone activations. Novel attack strategies routing prohibited reasoning through genuinely unprecedented feature combinations may not be captured by current probes. The Mahalanobis distance detector provides partial coverage for unknown attack surfaces, but it is not a complete solution. This is a known open problem in mechanistic interpretability research.

2. FSA Coverage Gap

The 94% recall target explicitly acknowledges a 6% gap. The safety case for any deployment must formally characterize which prohibited behaviors fall outside the FSA's state graph and provide a residual risk argument for each category. Deployment without this characterization would not be authorized.

3. Lagrangian Constraint Scalability

The dual ascent procedure scales with the number of constraint functions. At 120+ sub-categories across three prohibited domains this is manageable at current frontier model scales, but may require careful engineering at future parameter scales. Constraint function pruning via coverage analysis is an open research problem.

4. Exception Pathway Latency

The 72-hour human quorum is operationally challenging for time-sensitive research workflows. Reducing this latency without compromising the accountability guarantee—external audit coverage, TPM hash chain integrity, and quorum independence—is an open engineering problem. Any proposed modification must demonstrate preservation of all three accountability properties before implementation.

Conclusion and Next Steps

The Sovereign Sentinel Architecture represents a proposed approach to frontier AI safety that integrates mechanisms across five abstraction levels. While the individual components build on substantial prior work, the unified framework and latency-optimized design represent novel contributions to the AI safety literature.

Critically, this architecture has not been empirically validated at scale. The performance specifications represent engineering targets based on theoretical analysis and small-scale feasibility studies, not production benchmarks. Rigorous validation requires resources and infrastructure available only at Tier 1 research laboratories.

This proposal seeks collaborative partnership with an established AI safety team to:

30. Conduct proof-of-concept implementation of Axis 1 (CLR-CRAE) and Axis 2 (FSA-HI)
31. Validate certified robustness bounds through extensive adversarial evaluation
32. Benchmark actual latency overhead under realistic serving conditions

33. Identify failure modes and implementation challenges through rigorous red-teaming

The proposed 24-month roadmap (Phases 0-3) represents a realistic path to validating the architecture's core mechanisms before considering production deployment or extending to the full five-axis system.

We believe this approach—combining defense-in-depth across multiple abstraction levels with careful attention to production latency constraints—represents a promising direction for making frontier AI systems both safer and deployable at scale. However, we acknowledge that only rigorous empirical testing can determine whether these mechanisms deliver the proposed safety guarantees in practice.